

Initial Mechanical Design of a Simple Robotic Dog

Design Phase Technical Report

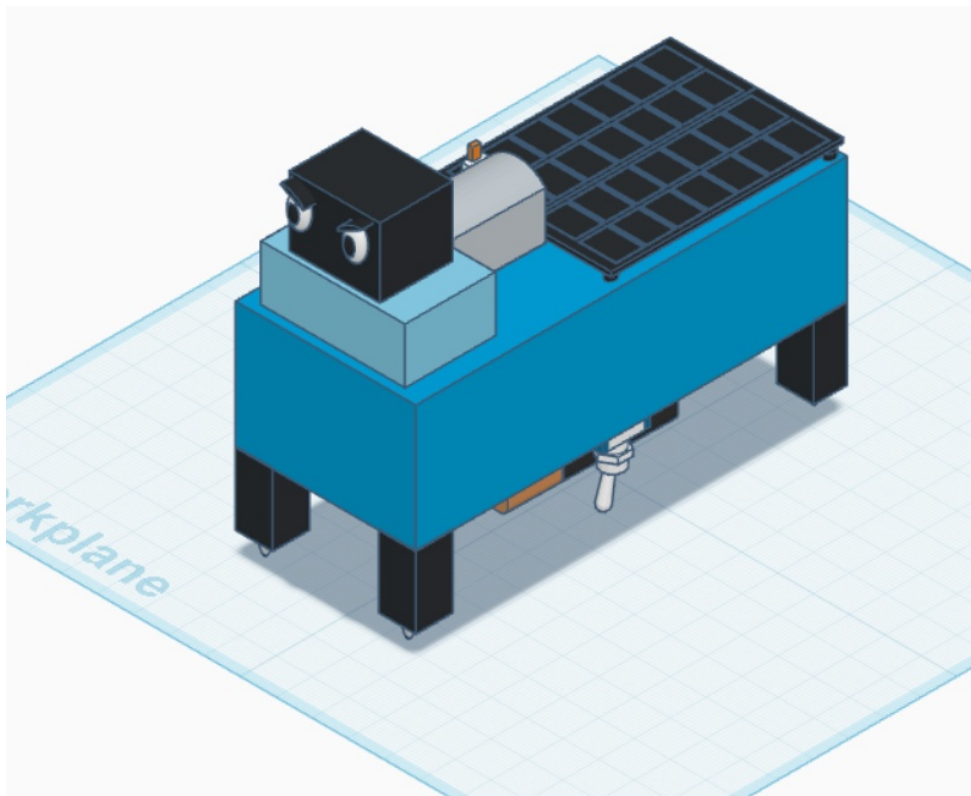


Figure 1: Main 3D isometric view of Faisal Albeshri robotic dog design.

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Training Program	Smart Methods Summer Training 2026

1. Executive Summary & Introduction

This report documents the preliminary mechanical design of a simple bio-inspired robotic dog. The objective is to understand the main mechanical principles that allow a four-legged robot to stand, carry components, and prepare for basic walking motion. The focus is not to create an advanced robot, but to study the structure, legs, joints, motor concept, initial torque demand, center of gravity, and expected mechanical issues.

The proposed design was modeled as a simple CAD concept using clear geometric shapes. It includes a rectangular chassis, a raised head block, four supporting legs, visual eyes, a DC motor representation, a 9V battery, an SPDT toggle switch, and two solar panels placed on the upper body.

2. Chassis Layout & Structural Geometry

The robot body is built around a rectangular main chassis. This shape was selected because it provides a stable base, a clear internal volume for components, and easy attachment points for the four legs. The upper deck carries lightweight components such as the solar panels and head assembly, while the heavier battery is positioned underneath the body to improve balance.

Part	Length	Width	Height	Design Purpose
Main Body / Chassis	120 mm	50 mm	35 mm	Primary support frame
Neck / Head Support	20 mm	20 mm	30 mm	Raises the head above body
Head	20 mm	20 mm	20 mm	Front visual block with eyes
Each Leg	10 mm	10 mm	20 mm	Basic vertical support

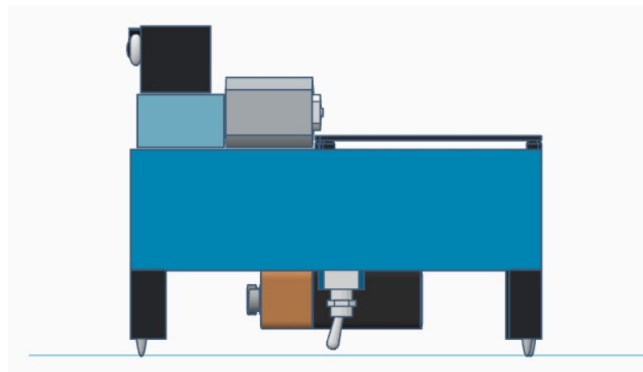


Figure 2: Side profile showing the long rectangular body and low battery position.

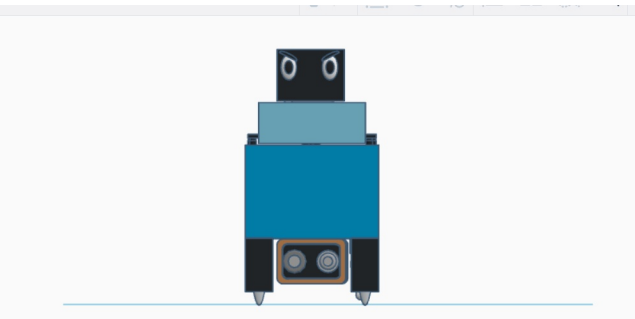


Figure 3: Front orthographic view showing head, eyes, legs, and lower components.

3. Head, Upper Deck & Visual Elements

The head is a small cube mounted at the front of the robot on a raised neck support. Two circular eye elements were added to make the design visually recognizable as a robotic dog. In an upgraded version, the eye area can be used to mount sensors such as ultrasonic distance sensors, infrared sensors, or a small camera.

Two solar panels are placed on the upper surface of the body. They are represented as lightweight top-mounted components. Their position keeps the design clean and visible without interfering with the lower support structure. The 9V battery and the SPDT switch are placed near the underside and central region, which helps maintain a lower center of gravity.

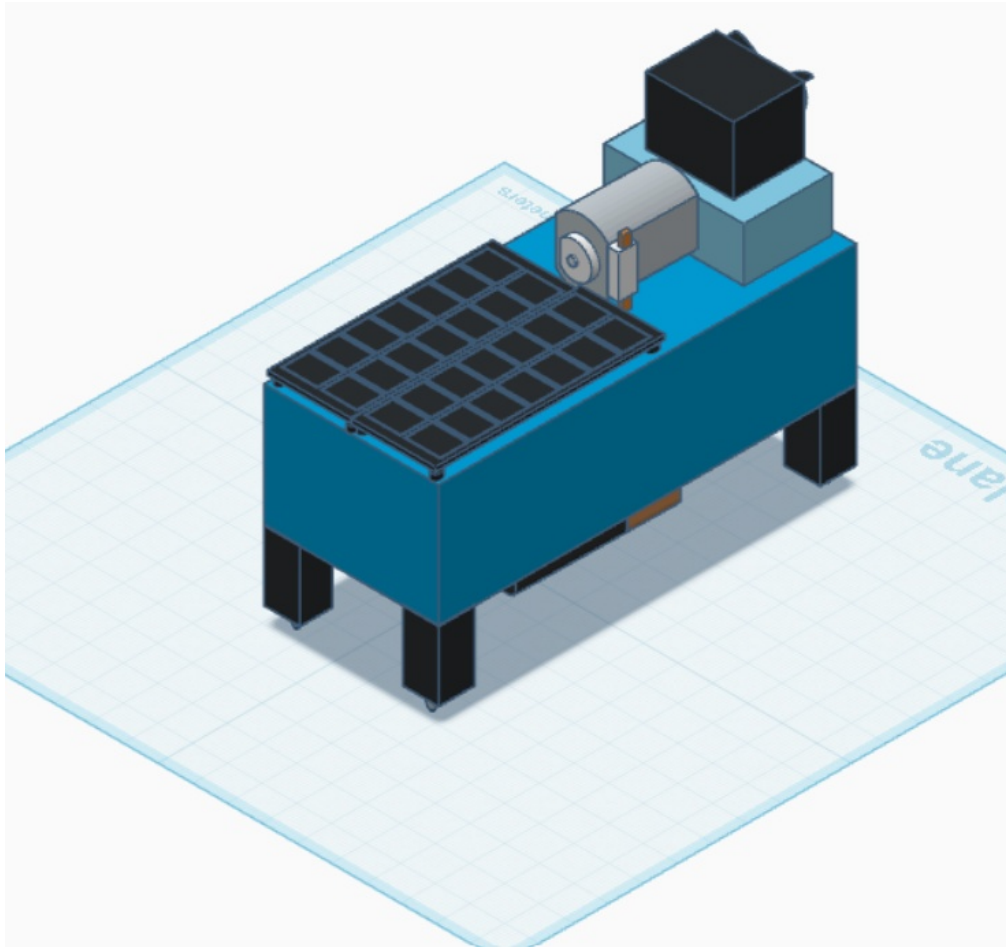


Figure 4: Upper deck arrangement with head, solar panels, DC motor representation, and chassis body.

4. Leg Architecture & Kinematic Degrees of Freedom

The robot uses four legs arranged near the corners of the rectangular chassis. This distribution creates a wide support base and improves static stability. The current model is intentionally simple, so each leg is considered to have one basic rotational walking joint. This supports the required understanding of simple standing and preliminary walking mechanics.

Limb Assembly	Joint Concept	Motion Profile	DOF
Front Right Leg	Single walking joint	Simple forward/backward leg swing	1 DOF
Front Left Leg	Single walking joint	Simple forward/backward leg swing	1 DOF
Rear Right Leg	Single walking joint	Simple forward/backward leg swing	1 DOF
Rear Left Leg	Single walking joint	Simple forward/backward leg swing	1 DOF
Total System DOF	-	4 legs x 1 DOF per leg	4 DOF

A future version can use two joints per leg, such as a hip joint and a knee joint. This would increase the design from 4 DOF to 8 DOF and allow smoother walking, but the current project target is a simple mechanical concept.

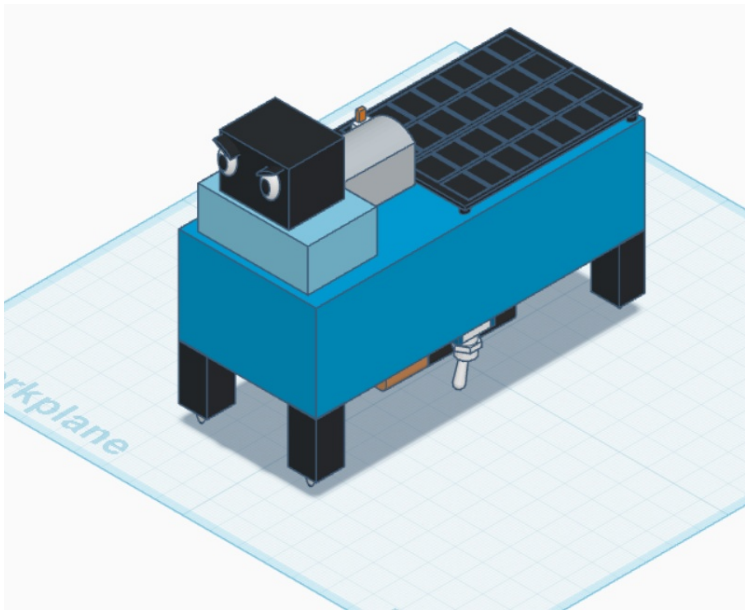


Figure 5: General view showing four-leg placement around the rectangular chassis.

5. Actuator Selection & Electrical Component Layout

The CAD model includes a DC motor rated for 3 V, 5 V, and 12 V operation. A DC motor is useful for demonstrating general mechanical motion because it is simple and easy to power. However, a normal DC motor does not directly control the exact angle of each leg, which is important for walking robots.

For the practical leg joints, the recommended actuator is a high-torque servo motor or a geared DC motor with feedback. A servo motor is preferred for a robotic dog because it can hold a specific angle and produce controlled leg movement. The 9V battery acts as the main power source, while the SPDT toggle switch provides a simple ON/OFF control for the system.

Component	Location in Design	Mechanical / Electrical Role
DC Motor 3-5-12 V	Top/front body region	Represents the motion source for the robot concept
9V Battery	Under the belly	Main power supply and heavy low-mounted component
SPDT Toggle Switch	Lower body area	Manual ON/OFF control
Two Solar Panels	Top of chassis	Auxiliary renewable-power concept and upper deck components
Eyes	Front of head	Visual robot-dog appearance; possible sensor position in future

6. Preliminary Static Torque Calculation

A basic torque estimation is required to check whether the selected joint actuator can hold one leg under static loading. Since the real mass has not been measured yet, the following calculation uses a reasonable preliminary assumption for a small CAD-based robotic dog.

Variable	Assumed Value	Meaning
m	0.80 kg	Estimated total robot mass
g	9.81 m/s ²	Acceleration due to gravity
Number of legs	4	Load is shared by four supports
r	0.03 m	Effective lever arm from joint to load point
FoS	2.5	Factor of safety for friction and motion

Total weight: $W = m \times g = 0.80 \times 9.81 = 7.85 \text{ N}$
 Force on one leg: $F_{\text{leg}} = W / 4 = 7.85 / 4 = 1.96 \text{ N}$
 Static torque: $T_{\text{static}} = F_{\text{leg}} \times r = 1.96 \times 0.03 = 0.0588 \text{ N.m}$
 Design torque with safety factor: $T_{\text{design}} = 0.0588 \times 2.5 = 0.147 \text{ N.m}$
 Servo metric conversion: $0.147 \text{ N.m} \times 10.197 = 1.50 \text{ kg.cm}$

Therefore, each joint actuator should provide at least 0.15 N.m, or approximately 1.5 kg.cm, for this preliminary static condition. For safer real movement, a higher-torque servo or geared motor should be selected because dynamic walking increases torque demand.

7. Stability Analysis & Center of Gravity

The stability of a quadruped robot depends on the position of its center of gravity relative to the support polygon formed by the four feet. In this model, the four legs are placed near the corners of the rectangular chassis. This creates a wide base that helps the robot resist tipping while standing.

The 9V battery is placed under the belly of the robot. This is an important mechanical choice because the battery is one of the heavier parts in the system. Placing it low and close to the center reduces the center of gravity height and improves static balance. The solar panels are placed on top, but they are thin and relatively light, so their effect on stability is limited.

Stability Rule: The vertical projection of the center of gravity should remain inside the rectangle formed by the four legs. If it moves outside this area, the robot may tip forward, backward, or sideways.

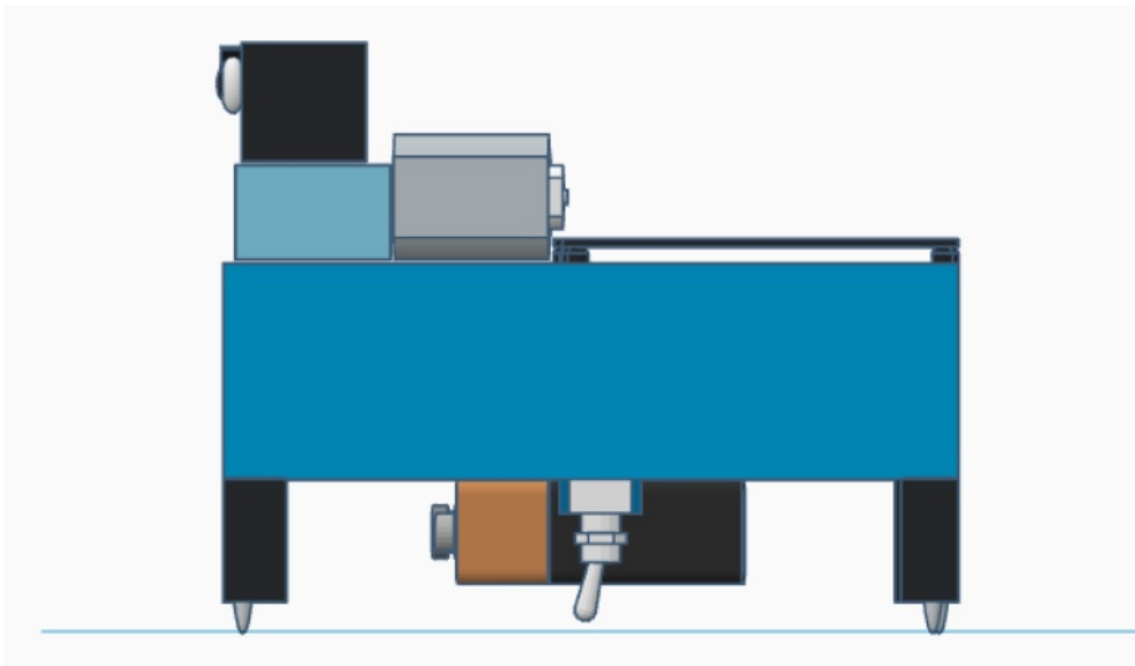


Figure 6: Side profile showing the lower battery location, which helps reduce the center of gravity.

8. Suggested Locomotion Strategy

For a simple four-legged robot, the safest initial walking method is a statically stable crawl gait. In this strategy, only one leg moves at a time while the other three legs remain on the ground. This keeps the center of gravity supported during each step and reduces the chance of tipping over.

- **Step 1:** Lift and move one front leg forward.
- **Step 2:** Move the opposite rear leg forward.
- **Step 3:** Move the second front leg forward.
- **Step 4:** Move the last rear leg forward to complete one cycle.

9. Expected Mechanical Problems

During a real physical build, several mechanical issues may appear. These problems should be considered before manufacturing or assembling the prototype.

Problem	Reason	Suggested Improvement
Weak leg support	The legs are thin compared with the main body.	Increase thickness or use stronger material.
Joint friction	Simple joints may rub against plastic surfaces.	Use smoother pivots, washers, or bushings.
Low motor torque	A small motor may not lift or hold the body.	Choose a geared motor or servo above the calculated torque.
Weight imbalance	Battery, motor, or panels may shift the center of gravity.	Keep heavy parts low and centered.
Limited walking motion	One DOF per leg creates basic movement only.	Add a knee joint in future versions.

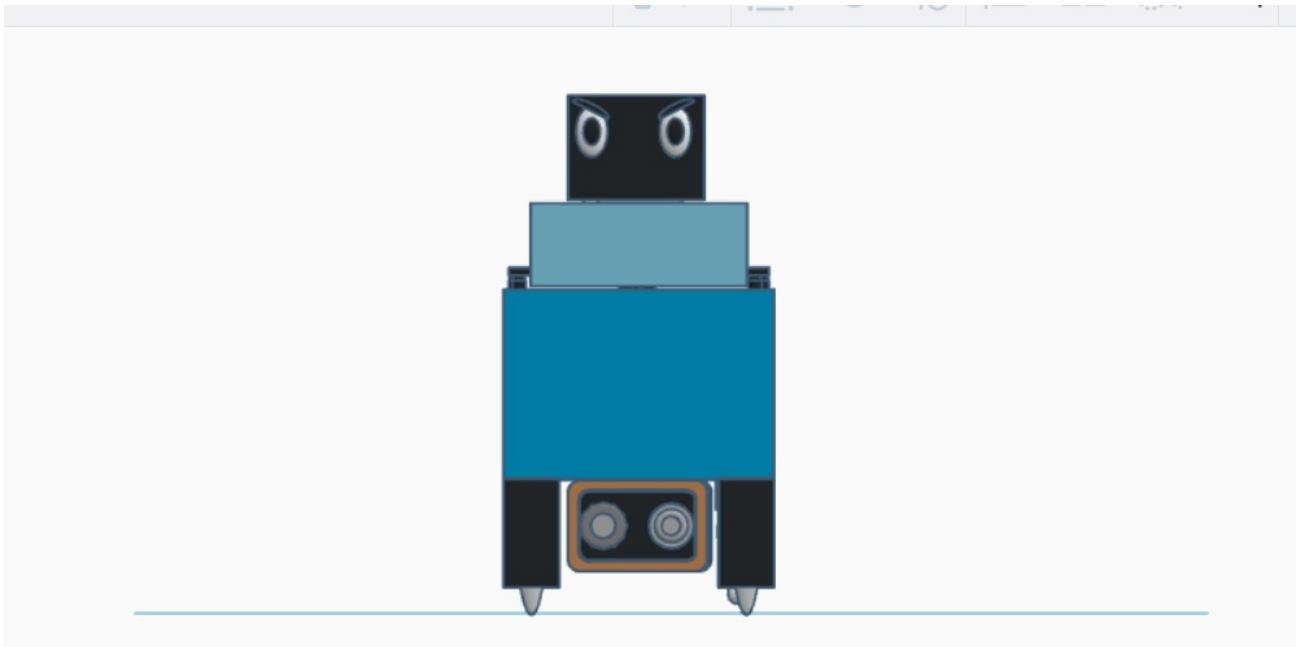


Figure 7: Front view highlighting symmetric leg placement and the lower component area.

10. Conclusion

This project presents a clear preliminary mechanical design for a simple robotic dog. The design uses a rectangular chassis, four supporting legs, a raised head, eye elements, a DC motor representation, a 9V battery, an SPDT toggle switch, and two solar panels. The structure is simple enough to meet the project target while still showing the basic mechanical ideas needed for a quadruped robot.

The initial analysis shows that the robot has 4 DOF, based on one simplified walking joint per leg. A preliminary torque calculation was completed for one joint, and the design torque requirement was estimated as approximately 0.15 N.m. The stability concept was improved by placing the heavier battery below the body and keeping the legs near the corners of the chassis.

Future improvements could include stronger legs, high-torque servo motors, two joints per leg, smoother pivots, sensor integration, and a more advanced walking sequence. Overall, the design successfully demonstrates the main mechanical principles required for a simple robotic dog to stand and prepare for basic movement.